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[Global E-Commerce Dataset \(+1M Records, 2024–2026\)](#)

# MINOR PROJECT REPORT :2

## Survey Data Analysis using K-Means Clustering

### Problem Statement

The objective of this project is to analyze a survey dataset and identify meaningful groups of similar data points using the K-Means clustering algorithm. Since the dataset does not contain predefined labels, an unsupervised learning approach is used to discover hidden patterns in the data.

### Dataset Description

- **Dataset Name:** Survey Dataset
- **Type:** Structured Dataset
- **Learning Type:** Unsupervised Learning
- **Features:** Multiple numerical and categorical attributes collected from survey responses.
- **Target Variable:** None

The dataset was cleaned and preprocessed before applying the clustering algorithm.

### Data Preprocessing Steps

The following preprocessing steps were performed:

- Imported the dataset.
- Checked missing values.
- Removed unnecessary columns (if any).

- Encoded categorical variables using Label Encoding.
  - Applied Feature Scaling using StandardScaler.
  - Prepared the dataset for clustering.
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## Exploratory Data Analysis (EDA)

The following EDA techniques were performed:

- Displayed first and last five rows.
- Checked dataset information using `info()`.
- Generated statistical summary using `describe()`.
- Checked missing values.
- Visualized data distributions.
- Examined relationships between important variables.

## Model Used

- **K-Means Clustering**
- K-Means is an unsupervised machine learning algorithm that groups similar data points into K different clusters based on their similarity.
- The optimal number of clusters was selected using the **Elbow Method**.

## Training Process

The following steps were followed:

1. Data preprocessing
2. Encoding
3. Feature Scaling
4. Elbow Method
5. Selection of optimal K
6. Applied K-Means Clustering
7. Assigned Cluster Labels
8. Analyzed Cluster Count
9. Visualized clusters using PCA

## Evaluation Metrics and Results

Since K-Means is an **unsupervised learning algorithm**, traditional classification metrics such as **Accuracy, Precision, Recall, F1-Score, and Confusion Matrix are not applicable**.

Instead, the clustering model was evaluated using:

- **Elbow Method** to determine the optimal number of clusters.
- **Cluster Distribution** to observe the number of records in each cluster.
- **PCA Visualization** to visually inspect the separation between clusters.

## Result

The K-Means algorithm successfully grouped the survey dataset into meaningful clusters. The visualization showed that similar data points were grouped together, indicating effective clustering.

## Conclusion

The survey dataset was successfully analyzed using the K-Means clustering algorithm. The data was preprocessed through encoding and feature scaling before applying the model. The Elbow Method helped determine the optimal number of clusters. K-Means successfully grouped similar data points into distinct clusters, and visualization using PCA provided a clear understanding of the clustering results. This project demonstrates that K-Means is an effective unsupervised learning technique for discovering hidden patterns in unlabeled data.

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## References

1. Scikit-learn Documentation
2. Python Documentation
3. Google Colab Documentation
4. NumPy Documentation
5. Pandas Documentation
6. Matplotlib Documentation

## ANSWER THE FOLLOWING

### 1. Semantic vs Instance vs Panoptic Segmentation

| Semantic Segmentation                     | Instance Segmentation                         | Panoptic Segmentation                               |
|-------------------------------------------|-----------------------------------------------|-----------------------------------------------------|
| Labels every pixel by class               | Separates different objects of the same class | Combines semantic and instance segmentation         |
| Objects of the same class share one label | Every object gets its own label               | Every pixel has both class and instance information |

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## 2. What is Fuzzy Logic and how is it different from Boolean Logic?

### Fuzzy Logic

Fuzzy Logic allows values between **0 and 1**, representing partial truth.

Example:

- Temperature = 0.8 Hot
- Temperature = 0.3 Cold

### Boolean Logic

Boolean Logic has only two values:

- True (1)
- False (0)

### Difference

| Fuzzy Logic           | Boolean Logic      |
|-----------------------|--------------------|
| Multiple truth values | Only True or False |

Handles  
uncertainty

Does not handle  
uncertainty

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**3. Suppose I give you a customer dataset containing Age, Annual Income, and Spending Score, but I don't provide any labels.**

**(1) Is this a supervised or unsupervised learning problem?**

**Answer:** Unsupervised Learning, because the dataset has no labels.

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**(2) Which clustering algorithm would you choose first and why?**

**Answer:** I would choose **K-Means Clustering** because it is simple, efficient, easy to implement, and works well for grouping similar customer data.

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**(3) What assumptions does that algorithm make about the data?**

**Answer:**

- The number of clusters (K) is known beforehand.
- Similar data points are close to each other.
- Clusters are approximately spherical.
- Features are properly scaled.
- Each data point belongs to only one cluster.